Defence practice

20 QUESTIONS | 20 MINUTES | 15 SINGLE + 5 MULTIPLE

Try the paper before reading the key on page 7. Questions 1-15: choose one answer; 16-20: select all that apply. Vertex and original edge IDs start at 0. This is ungraded practice.

01 DSU operation states

CHOOSE ONE

Start with seven singleton components. After each union below, record [components, largest_size]. Which sequence is correct?

```
n = 7
unions = [(1,4), (4,6), (1,6), (0,2)]
```

A [[6,2], [5,3], [4,3], [3,3]]
B [[6,2], [5,3], [5,3], [4,3]]
C [[6,2], [5,2], [5,2], [4,2]]
D [[6,2], [5,3], [5,4], [4,4]]

02 Whole parent array

CHOOSE ONE

What is the entire parent array after this path-compression loop?

```
int[] parent = {0, 0, 1, 2, 4, 4};
int root = 0, x = 3;
for (int u = x; u != root; ) {
    int next = parent[u];
    parent[u] = root;
    u = next;
}
```

A [0,0,1,0,4,4]
B [0,0,0,2,4,4]
C [0,0,0,0,0,0]
D [0,0,0,0,4,4]

03 Original edge IDs

CHOOSE ONE

For this directed graph, which original edge IDs leave vertex 2? Keep input order and include self-loops.

```
n = 4; vertex = 2
edges = [[2,0,7], [0,1,3], [2,2,-1],
         [1,2,4], [2,3,2]]
```

A [0,2,4]
B [0,3,4]
C [0,4]
D [1,2,3]

Defence practice

04 Kruskal trace

CHOOSE ONE

Run Kruskal in increasing weight order. Which original edge IDs are accepted, in acceptance order?

```
n = 5
edges = [[0,1,5], [1,2,1], [0,2,3],
         [3,4,-2], [2,2,-7]]
```

A [4,3,1,2]
B [3,1,0]
C [3,1,2]
D [1,2,3]

05 Prim heap candidates

CHOOSE ONE

A lazy Prim implementation has visited vertices {0,2}. Which edge is accepted next after discarding stale heap entries? Each tuple is (weight, from, to, edge_id).

```
heap candidates:
(6,0,1,0), (2,2,1,1), (1,2,0,2), (4,2,3,3)
```

A edge_id 2
B edge_id 1
C edge_id 3
D edge_id 0

06 DFS finish sequence

CHOOSE ONE

Start unvisited roots in increasing vertex order. Scan outgoing neighbours in input-row order. In which order do vertices finish?

```
n = 5
edges = [[0,2], [0,1], [2,3], [1,3]]
```

A [3,2,1,0,4]
B [0,2,3,1,4]
C [3,1,2,0,4]
D [4,0,1,2,3]

07 SCC membership

CHOOSE ONE

Which component-label array represents exactly the strongly connected components? Labels are arbitrary nonnegative integers.

```
n = 6
edges = [[0,2], [2,0], [2,1], [1,3],
         [3,1], [3,4], [4,4]]
```

A [8,8,8,8,9,12]
B [8,4,8,9,9,12]
C [8,4,8,4,12,12]
D [8,4,8,4,9,12]

Defence practice

08 Topological order

CHOOSE ONE

Which sequence is a valid topological order of all five vertices?

```
edges = [[0,3], [1,3], [1,2], [2,4]]
```

A [0,3,1,2,4]
B [1,0,4,2,3]
C [1,2,0,4,3]
D [2,1,0,3,4]

09 DAG shortest distances

CHOOSE ONE

What is the distance array? The graph is directed; null denotes an unreachable vertex.

```
n = 6; source = 1
edges = [[1,0,4], [1,2,7], [0,2,-6],
         [2,3,3], [0,3,5], [4,5,-8]]
```

A [4,0,7,9,null,null]
B [4,0,-2,1,null,null]
C [4,0,-2,1,0,-8]
D [0,4,-6,-3,null,null]

10 Route reconstruction

CHOOSE ONE

Using these original edge IDs, reconstruct the route from source 1 to target 3. Which pair gives the vertex route and its edge IDs?

```
edges = [[1,0,4], [1,2,7], [0,2,-6],
         [2,3,3], [0,3,5], [4,5,-8]]
parent_edge = [0,-1,2,3,-1,-1]
```

A vertices [1,0,2,3]; edge IDs [0,2,3]
B vertices [1,2,3]; edge IDs [1,3]
C vertices [3,2,0,1]; edge IDs [3,2,0]
D vertices [1,0,2,3]; edge IDs [1,0,2]

11 Minimum spanning forest

CHOOSE ONE

Which triple is (minimum cost, component count, selected-edge count)? The graph is undirected.

```
n = 6
edges = [[0,1,-3], [1,2,4], [0,2,2],
         [3,4,6], [3,4,1], [5,5,-20]]
```

A (-20,3,3)
B (0,2,4)
C (2,3,3)
D (0,3,3)

Defence practice

12 Condensation graph

CHOOSE ONE

For the given valid SCC labels, which condensation array is correct? Pair order is irrelevant; duplicate pairs must be omitted.

```
component = [2,2,8,8,5]
edges = [[0,1], [1,2], [0,3], [2,3],
         [3,4], [2,4], [4,4]]
```

A [[2,8],[8,5],[5,5]]
B [[2,8],[8,5]]
C [[2,8],[2,8],[8,5],[8,5]]
D [[8,2],[5,8]]

13 Union by size

CHOOSE ONE

Call union(1,4), attaching the smaller root below the larger root. Initially there are three components. Which triple is (parent[0], size[2], components) afterward? Sizes at non-roots are ignored.

```
parent = [0,0,2,2,2,5]
size   = [2,1,3,1,1,1]
```

A (0,5,2)
B (2,3,2)
C (2,5,2)
D (2,5,1)

14 Cycles outside source reachability

CHOOSE ONE

Which result satisfies the Assignment 3 DAG contract?

```
n = 5; source = 0
edges = [[0,1,2], [2,3,1], [3,2,1]]
```

A acyclic=false; distance=[]; parent_edge=[]
B acyclic=true; distance=[0,2,null,null,null]; parent_edge=[-1,0,-1,-1,-1]
C acyclic=false; distance=[0,2,null,null,null]; parent_edge=[-1,0,-1,-1,-1]
D acyclic=true; distance=[]; parent_edge=[]

15 Benchmark medians

CHOOSE ONE

Both methods were timed on the same graph with the same timing boundary. Which pair gives (Kruskal median, Prim median), in milliseconds?

```
Kruskal: [9,4,6] ms
Prim:    [5,8,7] ms
```

A (4,5)
B (9,8)
C (6.33,6.67)
D (6,7)

Defence practice

16 Alternative optimal forests

SELECT ALL THAT APPLY

Select all edge-ID sets that are minimum spanning forests. IDs inside a set are unordered. The graph is undirected.

```
n = 5
edges = [[0,1,2], [1,2,2], [0,2,2],
         [2,3,5], [1,3,9]]
```

A [0,1,3]
B [0,1,4]
C [0,2,3]
D [0,1,2]
E [1,2,3]

17 DFS active edges

SELECT ALL THAT APPLY

Use three-state DFS. Start roots in increasing vertex order and scan neighbours in input-row order. Select all edges that lead to an active vertex when examined.

```
n = 5
edges = [[0,1], [1,2], [2,0], [1,3],
         [3,1], [3,4], [4,4]]
```

A ID 0: 0 -> 1
B ID 2: 2 -> 0
C ID 4: 3 -> 1
D ID 5: 3 -> 4
E ID 6: 4 -> 4

18 Topological witnesses

SELECT ALL THAT APPLY

Select all valid topological orders. Each must contain every vertex exactly once.

```
n = 4
edges = [[0,2], [1,2], [1,3]]
```

A [0,2,1,3]
B [0,1,2,3]
C [1,2,0,3]
D [3,1,0,2]
E [1,0,3,2]

Defence practice

19 Shortest-path parent witnesses

SELECT ALL THAT APPLY

Select all valid parent_edge arrays for shortest paths from source 0. Entries store original incoming edge IDs.

```
n = 4; source = 0
edges = [[0,1,3], [0,1,3], [1,2,-5],
         [0,2,0], [2,3,4], [1,3,2]]
```

A [-1,0,2,4]
B [-1,0,3,4]
C [-1,2,2,4]
D [-1,1,2,4]
E [-1,0,2,5]

20 Test cases for a serialization defect

SELECT ALL THAT APPLY

A DAG solver computes correct finite distances but serializes every distance equal to -1 as null. Select all requests that expose this defect. All requests have n=3 and source=0; only edges are shown.

A [[0,1,2],[1,2,-3]]
B [[0,1,4],[1,2,-2]]
C [[0,1,-1]]
D [[1,2,-1]]
E [[0,1,-2],[1,2,1]]

Answers & explanations

01 DSU operation states — B

The first two unions create {1,4,6}. The third joins vertices already in that component, so it changes neither count nor size. The last creates {0,2}. The remaining singleton vertices are 3 and 5.

02 Whole parent array — D

The old path is 3 -> 2 -> 1 -> 0. Saving next before assignment lets the loop visit the whole old path. Vertices 1, 2 and 3 point to root 0; the separate component {4,5} is unchanged.

03 Original edge IDs — A

An outgoing edge has 2 in its first column. Rows 0, 2 and 4 satisfy this test. Row 3 enters vertex 2; row 2 is a self-loop and is included once. IDs refer to original rows.

04 Kruskal trace — C

The sorted order is 4, 3, 1, 2, 0. Edge 4 is a self-loop and is rejected. Edges 3, 1 and 2 connect different components; edge 0 then closes a cycle. Cost is -2 + 1 + 3 = 2, with two components.

05 Prim heap candidates — B

The lightest entry, edge 2, ends at already visited vertex 0 and is stale. The next lightest edge reaching an unvisited vertex is edge 1, with weight 2 and destination 1.

06 DFS finish sequence — A

DFS from 0 enters 2, then 3. Vertex 3 finishes before 2. Next, DFS visits 1, whose neighbour 3 is already finished. Finally 0 finishes. The outer loop later visits isolated vertex 4. This is finish order, before reversal.

07 SCC membership — D

Mutual reachability gives {0,2}, {1,3}, {4}, {5}. The edges from one group to the next do not provide a return path, so they do not merge the groups. The self-loop at 4 does not connect it to another vertex.

Answers & explanations

08 Topological order — C

In [1,2,0,4,3], every edge goes from an earlier vertex to a later one. The other choices violate 1 -> 3, 2 -> 4 or 1 -> 2, respectively. The assignment accepts any valid order, not only one DFS output.

09 DAG shortest distances — B

Vertex 0 costs 4. Reaching 2 through 0 costs 4 - 6 = -2, improving on 7. Reaching 3 through 2 costs -2 + 3 = 1, improving on 9. Vertices 4 and 5 are unreachable; their edge must not be relaxed from an invented finite distance.

10 Route reconstruction — A

From target 3, parent edge 3 leads back to 2; edge 2 leads back to 0; edge 0 leads back to source 1. Reverse the collected sequence. The selected edge weights sum to 4 - 6 + 3 = 1. Parent entries are edge IDs, not predecessor vertices.

11 Minimum spanning forest — D

For {0,1,2}, select weights -3 and 2, giving -1. For {3,4}, select the cheaper parallel edge of weight 1. Vertex 5 is isolated for connectivity purposes; its loop cannot enter a forest. Total cost is 0, with three components and n-c = 3 edges.

12 Condensation graph — B

Remove edges whose endpoints have equal labels. Map the other edges to label pairs and deduplicate: 1 -> 2 and 0 -> 3 both become (2,8); 3 -> 4 and 2 -> 4 both become (8,5). Keep the original directions.

13 Union by size — C

find(1)=0 and find(4)=2. Root 0 has size 2 and root 2 has size 3, so parent[0] becomes 2. Add the component sizes at root 2 to obtain 5. One successful merge decreases the component count from 3 to 2.

14 Cycles outside source reachability — A

Vertices 2 and 3 form a directed cycle even though the source cannot reach them. The required cycle check covers the entire graph. Any directed cycle requires acyclic=false and both arrays empty.

Answers & explanations

15 Benchmark medians — D

Sort each sample: [4,6,9] and [5,7,8]. The middle values are 6 and 7. Kruskal has the lower median for this measured graph and environment; that observation alone does not establish which method is faster on all graphs.

16 Alternative optimal forests — A, C, E

Vertex 4 is isolated. Any two edges from the equal-weight triangle connect {0,1,2} for cost 4. Add edge 3 to reach vertex 3 for total cost 9. Using edge 4 costs more; using all three triangle edges creates a cycle and leaves vertex 3 disconnected.

17 DFS active edges — B, C, E

DFS reaches 0 -> 1 -> 2, where edge 2 points back to active 0. After 2 finishes, DFS enters 3; edge 4 points to active 1. At vertex 4, the self-loop points to active 4. Edges 0 and 5 discover unvisited vertices, so they are tree edges.

18 Topological witnesses — B, E

Both 0 and 1 must precede 2, and 1 must precede 3. The second and fifth choices meet every condition. The first places 2 before 1; the third places 2 before 0; the fourth places 3 before 1.

19 Shortest-path parent witnesses — A, D

Distances are [0,3,-2,2]. Either parallel edge 0 or 1 may reach vertex 1. Edge 2 is needed to reach 2 at cost -2, and edge 4 reaches 3 at cost 2. Edge 3 gives the wrong distance to 2; edge 2 ends at 2, not 1; edge 5 reaches 3 at cost 5.

20 Test cases for a serialization defect — A, C, E

The first and fifth requests produce distance -1 at vertex 2; the third produces -1 at vertex 1. The second has only 0, 4 and 2 as finite distances. In the fourth, vertices 1 and 2 are unreachable, so null is already the correct output for them.

Assignment specification / Interactive lesson